



Islamic University of Najaf
College of Technical Engineering
Department of Computer Technical Engineering
2026-2027

Information Theory and Coding
Lecture 1

Fundamentals of Information Theory

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Course Communication

Scan the QR code or use the link to join the class:

<https://classroom.google.com/c/ODg1OTAzNjgzODU1?cjc=bysyzqop>



Lecture Description

Lecture Title	Fundamentals of Information Theory
Lecture location	Islamic University of Najaf / College of Technical Engineering / Department of Computer Technical Engineering
Target audience	4th Stage Students\ Department of Computer Technical Engineering
Year	2026-2027
Educational methods and techniques	Computer – Electronic screen – Classroom - Board
Assessment Methods	Activities – Tasks
Time	2 hours

Lecture Objectives

After this lecture, the student will be able to:

- ✓ Define probability as a mathematical tool for uncertainty.
- ✓ Define random variables.
- ✓ Distinguish between discrete and continuous random variables.
- ✓ Define sample spaces, events, and basic probability rules.



THINK ABOUT IT...

I already
know what
it says...



If you receive a message and you already know its content with **100% certainty**, does that message still carry **information**?

Why or why not?



*Share
your thoughts!*



Probability is a measure of how likely an event is to occur.

For example:

- ☐ Today there is a 60% chance of rain.
- ☐ The chances of winning the lottery are a million to one.

Probability in Communication Systems

Probability studies randomness and uncertainty.

Signal Random process

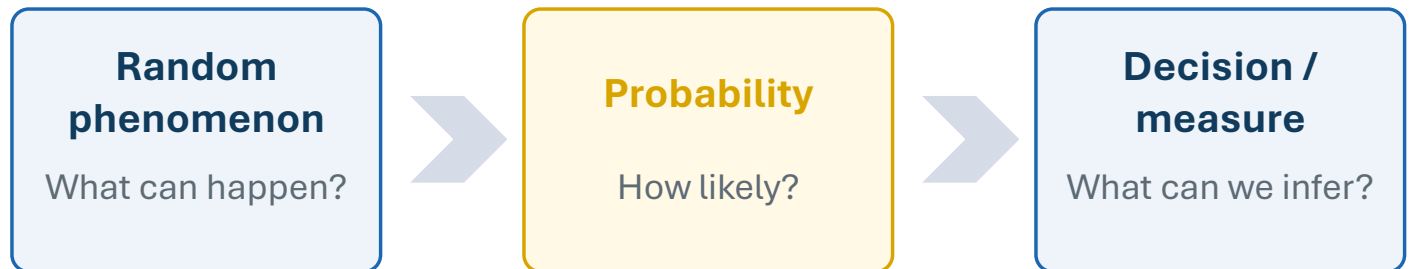
Message Random / uncertain;
no uncertainty → no
new information

Interference Random

Noise Random process

**Delay /
Phase /
Fading** Often modeled
probabilistically

Uncertainty → Probability Model → Quantitative Analysis



$$0 \leq P(A) \leq 1$$

Variables & Random Variables

A variable is any characteristics, number, or quantity that can be measured or counted. There are different ways variables can be described according to the ways they can be studied, measured, and presented.

Random Variables a is a process whose outcome is uncertain. A random variable, usually written X , is a variable whose possible values are numerical outcomes of a random phenomenon. There are two types of random variables, discrete and continuous.

Random Variables Types

Discrete Random Variable

Takes a countable number of distinct values.

Examples:

- Number of children in a family
- Number of defective bulbs in a box
- Die result: 1, 2, 3, 4, 5, 6

Continuous Random Variable

Can take infinitely many possible values over an interval.

Examples:

- Time taken to complete a task
- Measured amplitude
- Temperature or other physical measurements

All random variables have a cumulative distribution function (CDF).

Discrete Random Variable

$$P(A) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$$



Experiment: Roll one fair die

$S = \{1, 2, 3, 4, 5, 6\}$

X = number shown on the die

$X \in \{1, 2, 3, 4, 5, 6\}$

$P(X=1)=P(X=2)=\dots=P(X=6)=1/6$

.....

$A = \{\text{even number}\} = \{2, 4, 6\}$

$P(A) = 3/6 = 1/2$

Discrete Cumulative Distribution Function

The CDF gives the probability that X is less than or equal to a specified value.

$$F(x) = P(X \leq x)$$

Given:

Outcome	1	2	3	4
Probability	0.1	0.3	0.4	0.2

Cumulative values

$$F(1) = 0.1 \quad 0.1$$

$$F(2) = 0.4 \quad 0.1 + 0.3$$

$$F(3) = 0.8 \quad 0.1 + 0.3 + 0.4$$

$$F(4) = 1 \quad 0.1 + 0.3 + 0.4 + 0.2$$

The CDF is nondecreasing and reaches 1 when all probability has accumulated.

Activity

Given the following probability distribution:

Outcome x	1	2	3	4
$P(X = x)$	0.2	0.1	0.4	0.3

1. Calculate: $F(1), F(2), F(3), F(4)$
2. Find: $P(x \leq 2)$
3. Find: $P(x \leq 3)$

Continuous Random Variables

A continuous random variable is described by a probability density function (PDF).

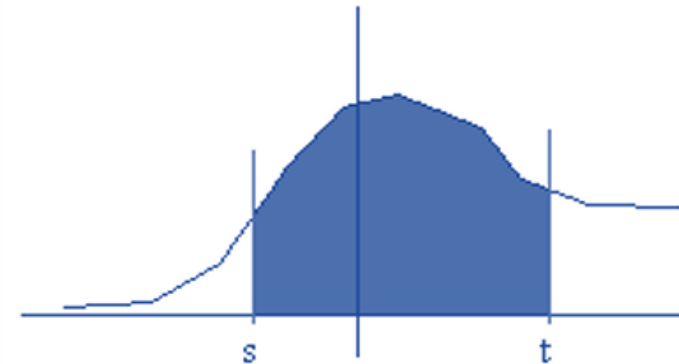
$$\int_{-\infty}^{\infty} f(x) dx = 1$$

Density curve requirements:

$f(x) \geq 0$ for all x

Total area under the curve = 1

Probability density sketch



Area = 1

Continuous CDF and Interval Probability

For a continuous variable, the CDF is obtained by integrating the PDF.

$$F(t) = P(X \leq t) = \int_s^t f(x) dx$$

$$P(s \leq X \leq t) = F(t) - F(s)$$

Example: $f(x) = c x, 0 \leq x \leq 1$

Normalize the density:

$$\begin{aligned} & \int_0^1 cx \, dx \\ &= c/2 \\ &= 1, \text{ Therefore: } c = 2 \end{aligned}$$

Sample Spaces and Events

A sample space contains all possible outcomes of an experiment.



Sample Space (S or Ω)

The set of all possible outcomes.

Coin toss: $S = \{H, T\}$

Sample Point

One individual outcome in the sample space.

Example: H or T

Event

A subset of the sample space.

Example: even outcome on a die

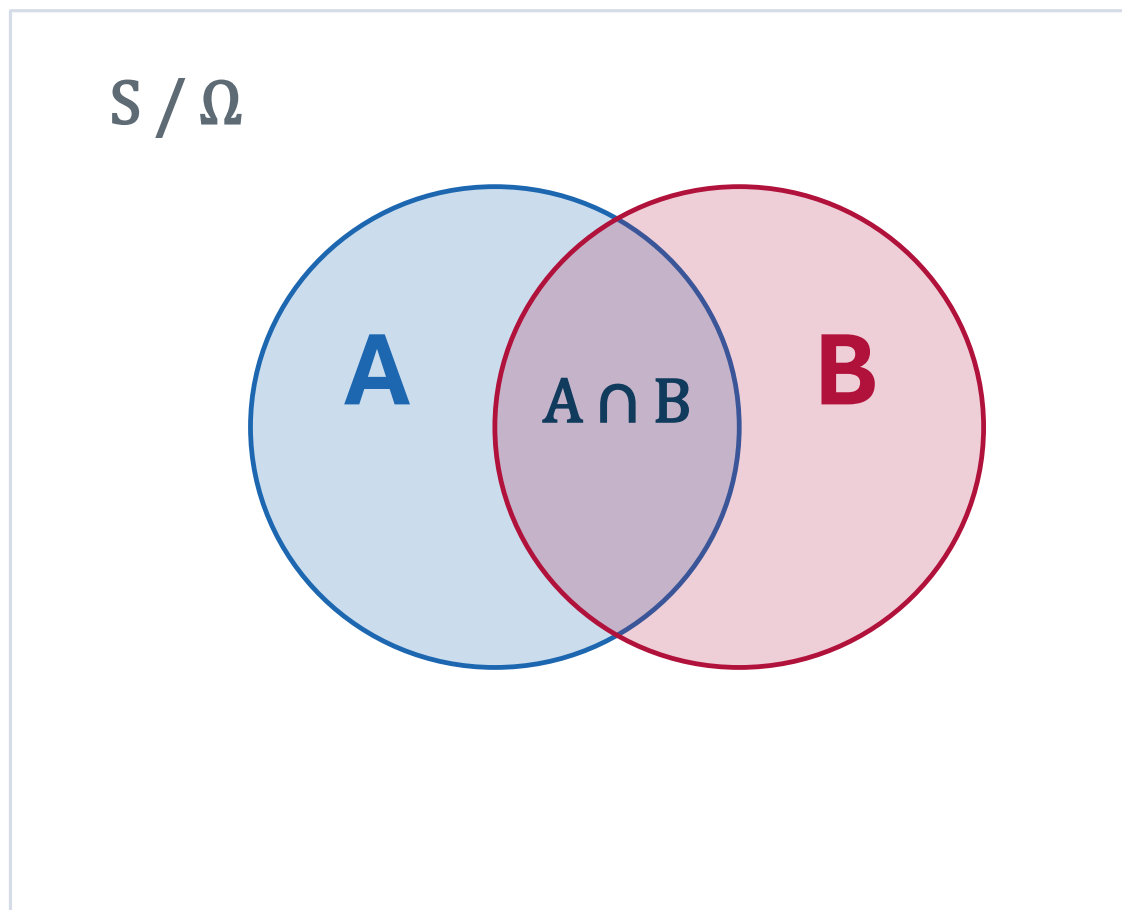
Example: Toss a coin three times

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

Event A: at least two heads $A = \{HHH, HHT, HTH, THH\}$

Venn Diagrams and Set Operations

Venn diagrams visualize logical relationships between events.



Useful Set Laws

Commutative

$$A \cup B = B \cup A$$

$$A \cap B = B \cap A$$

Associative

$$(A \cup B) \cup C = A \cup (B \cup C)$$

$$(A \cap B) \cap C = A \cap (B \cap C)$$

Distributive

$$(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$$

Probability of an Event

For equally likely outcomes, probability is the fraction of favorable outcomes.

$$P(A) = \text{Number of outcomes in } A / \text{Number of outcomes in } S$$

Range

$$0 \leq P(A) \leq 1$$

Impossible event

$$P(\emptyset) = 0$$

Certain event

$$P(S) = 1$$

Partition

If $A_1, \dots, A_n$ are mutually exclusive and cover A , then $P(A) = \sum P(A_i)$

Good solution steps: define the experiment → sample space → event → probability.

Example: Three Coin Tosses

Find the probability of obtaining at least two heads.



$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

Step 1 — Event

At least two heads:

$$A = \{HHH, HHT, HTH, THH\}$$

Step 2 — Count

Favorable outcomes = 4

Total outcomes = 8

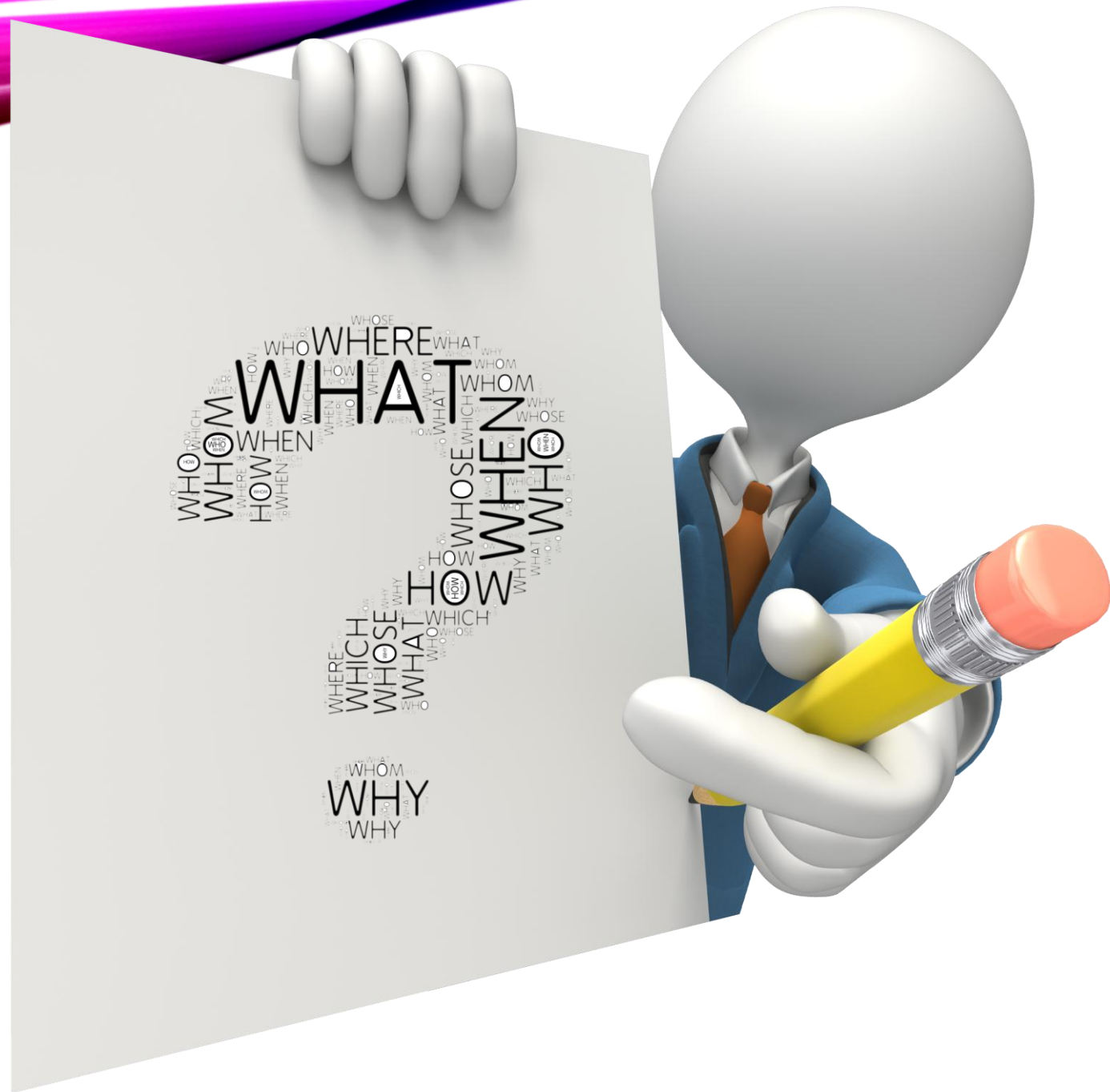
Step 3 — Probability

$$P(A) = 4/8 = 1/2$$

$$P(A) = 0.5$$

Homework

Q1: A fair die is rolled twice. Find the probability that the sum of the two rolls is 7.



Thank you

